

ABSTRACT

The aim of the "Night Vision Motion Detection" project is to combine automatic alarm triggers, real-time motion detection, and night vision capabilities to improve security systems. For this project, a dedicated camera that can transmit live video feeds to continuously monitor locations, even in low light. The film Strong open-source software called OpenCV is used to process the camera's frame captures library for computer vision. This processing uses sophisticated algorithms that are capable of distinguish between background noise and human movement to ensure precise motion while reducing the number of false alarms caused by non-human activity. The device plays a recorded audio clip in response to human motion detection, warning residents or security staff of possible dangers. The undertaking integrates the most recent developments in audio and computer vision technologies to provide a thorough and efficient monitoring solution that works for homes, businesses, as well as public spaces. Using precise motion detection and night vision approaches, this initiative claims to provide strong security, particularly in dimly lit areas or circumstances during the night, providing security and comfort.